

Unit 4, Lesson 2: Rational Numbers and the Number Line – Part 2

Benchmarks

MA.6.NSO.1.1 Extend previous understanding of numbers to define rational numbers. Plot, order, and compare rational numbers.

MA.6.NSO.1.2 Given a mathematical or real-world context, represent quantities that have opposite direction using rational numbers. Compare them on a number line and explain the meaning of zero within its context.

Learning Targets

- ☐ I can determine how far a number and its opposite are from zero.
- ☐ I can represent a number and its opposite on a number line.
- ☐ I can compare a quantity and its opposite on a horizontal or vertical line and explain the meaning of zero within the context.

4.2.1 Warm-Up: Word Pairs

Up	Down
Left	Right
Clockwise	Counterclockwise
Increase	Decrease
Ebb	Flow
West	East
North	South
Gain	Lose

1. What do you notice about the pairs of words above?
2. How do you think this may be used in the study of mathematics?

4.2.2 Activity: Folded Number Line

On a separate sheet of paper, create a number line with integers from -10 to 10 to answer the following questions. Fold the number line so that the vertical crease goes through 0 and the two sides of the number line match up perfectly.

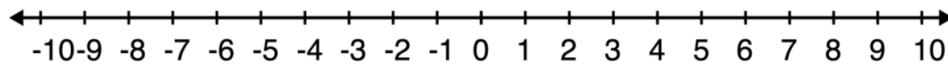
1. When the number line is folded, which number is the same distance away from 0 as the number 4 ?
2. When the number line is folded, which number is the same distance away from 0 as the number -7 ?
3. When the number line is folded, what do you notice about the numbers that are the same distance away from 0 ?
4. Find another pair of numbers that are the same distance away from 0 .
5. Two numbers that are the same distance from zero on the number line are called _____.

4.2.3 Collaboration: Opposites

With your partner, choose values for W and F . The value for W can be any rational number between -8 and 0 that is not an integer. The value for F can be any rational number between 0 and 8 that is not an integer.

The value for W is _____. The value for F is _____.

1. Plot and label both numbers on the number line.

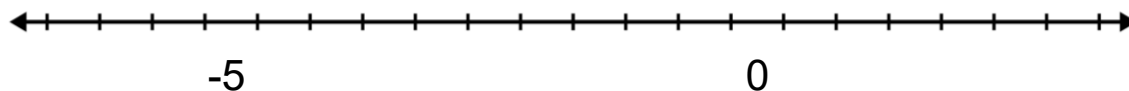


2. Plot the opposite of W . Label the point X .
3. What is the value of point X ?
4. How far is point X from 0 ?
5. Plot the opposite of F . Label the point G .
6. What is the value of point G ?
7. How far is point G from 0 ?
8. Noah said, "The expression $-a$ can be read 'negative a ' or 'the opposite of a '. So, if a is a rational number, $-a$ will always be a negative number."

Use points F , G , W , and X to explain to Noah that the opposite of a value could be positive.

4.2.4 Try It: Opposites

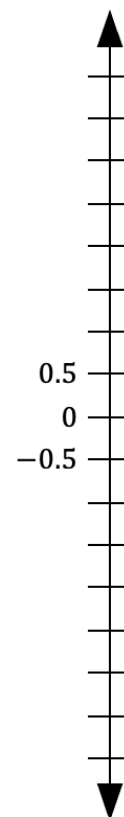
1. Use the number line below to answer the following problems.



- a. Label the tick marks on the number line. Then plot the value $\frac{5}{4}$ and its opposite on the number line.
- b. Which value is larger, $\frac{5}{4}$ or its opposite?

2. Use the vertical number line to answer the following problems.

- a. Label the tick marks on the number line. Plot and label the value -3.8 and its opposite on the number line.
- b. Which value is larger, -3.8 or its opposite?

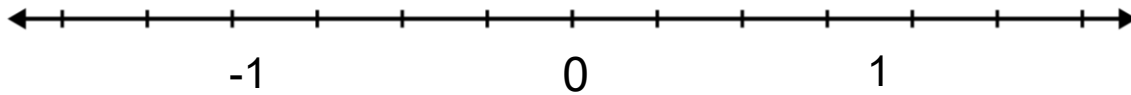


4.2.5 Guided Instruction: Comparing a Quantity and Its Opposite

Plot each pair of values on the number line. Choose an answer to correctly fill in the blank.

1.

- a. Plot -0.25 and 0.25 .

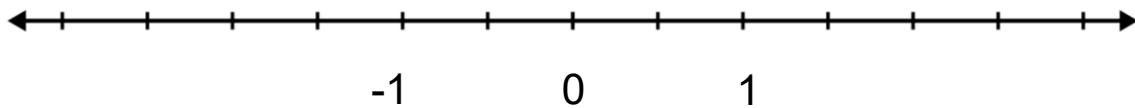


- b. The value -0.25 is ☐ greater ☐ less than 0.25 .

- c. -0.25 ☐ $>$ ☐ $<$ 0.25

2.

- a. Plot $2\frac{2}{5}$ and $-2\frac{2}{5}$.



- b. The value $2\frac{2}{5}$ is ☐ greater ☐ less than $-2\frac{2}{5}$.

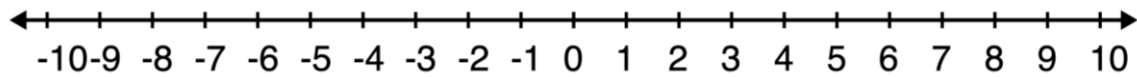
- c. $2\frac{2}{5}$ ☐ $>$ ☐ $<$ $-2\frac{2}{5}$

3. In each comparison, the number on a horizontal number line that is farther to the ☐ right ☐ left is the greater value.

4. When comparing opposites, the ☐ positive ☐ negative value will always be the greater value.

4.2.6 Your Turn!

1. Plot and label each value and its opposite on the number line: 9.5, -2 , 4.2, $-7\frac{3}{4}$.



2. Pairs of opposite numbers are ☐ the same
☐ a different distance from zero on a number line.
3. Complete each comparison statement for the pairs of numbers you plotted in problem #1. For each statement, use $<$ or $>$ in the first blank and the opposite of the given number in the second blank.
- a. 9.5 _____
- b. -2 _____
- c. 4.2 _____
- d. $-7\frac{3}{4}$ _____

4.2.7 Wrap-Up: Cool Down

1. What is the relationship between a number and its opposite? Use the words **number line** and **zero** in your explanation.